

LexAI: A Reliability-Constrained Retrieval-Augmented Generation System for Indian Legal Research in the Post-BNS Transition Era

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Abstract—Large language models deployed for legal research exhibit two critical failure modes in the Indian jurisdiction: citation hallucination and statutory obsolescence arising from the 2023 legislative transition that replaced the Indian Penal Code with the Bharatiya Nyaya Sanhita. We present LEXAI, a reliability-constrained retrieval-augmented generation system that addresses both failures through a seven-step retrieval pipeline with confidence-threshold abstention and a learned statutory transition classifier achieving 98.7 % F1 on held-out test data—outperforming a manually curated rule-based baseline (F1 = 0.704) by correctly handling 60 previously undetected transition cases. On a 393-query lawyer-verified benchmark spanning seven query categories and four legal domains, LEXAI achieves 95.5 % citation accuracy compared to 84.1 % for zero-shot LLM and 61.6 % for naïve RAG. The transition module reduces outdated-law citations from 82.3 % to 32.6 %, a 60 % relative reduction. Answer completeness reaches 72.5 versus 65.0 for naïve RAG. We release LEXEVAL-INDIA-393, the first lawyer-verified benchmark dataset for Indian legal QA spanning criminal, civil, corporate, and family law in the post-BNS transition period.

Index Terms—retrieval-augmented generation, legal NLP, Indian law, statutory transition, hallucination, BNS, IPC, cross-domain

I. INTRODUCTION

On 1 July 2024, three statutes that had governed Indian criminal law for more than 160 years ceased to have effect. The Bharatiya Nyaya Sanhita (BNS), Bharatiya Nagarik Suraksha Sanhita (BNSS), and Bharatiya Sakshya Adhinyam (BSA) replaced the Indian Penal Code (IPC), Code of Criminal Procedure (CrPC), and Indian Evidence Act (IEA) respectively [1]. For practitioners, the administrative transition was orderly; for AI-assisted legal research tools, it was quietly catastrophic. Any system trained or indexed before that date encodes superseded law as current law—with no indication to the user that anything has changed.

The practical stakes are not abstract. Junior advocates handling high volumes of routine criminal matters increasingly rely on AI tools for preliminary research [2]. A system that confidently cites IPC Section 302 when the applicable provision is now BNS Section 101 does not merely produce a wrong answer: it produces one that exposes the practitioner to procedural objections and erodes client trust—a professionally consequential error presented as authoritative guidance.

Retrieval-augmented generation (RAG) has become the dominant approach for knowledge-intensive legal QA [3], [4], grounding LLM responses in retrieved documents rather than relying on parametric memory. The appeal is clear, but standard RAG architectures are ill-equipped to handle temporal legislative discontinuities. A retrieval index can contain both IPC and BNS documents without any mechanism to flag supersession, leaving the model free to blend old and new law in ways that are difficult to detect.

This paper describes LEXAI, a RAG system designed with statutory reliability as a first-order constraint. Its central technical contribution is a *learned statutory transition classifier*—a fine-tuned sentence encoder paired with a logistic regression head—trained to detect references to superseded legislation within retrieved context and to enrich LLM input with the corresponding replacement provisions. Where a manually curated dictionary fails on abbreviated forms (*u/s*), vernacular shorthand, and less-common section references, the learned classifier generalises over sentence embeddings and recovers 60 such cases that the rule-based baseline misses entirely.

To evaluate the system rigorously, we constructed LEXEVAL-INDIA-393, a lawyer-verified benchmark dataset extending across four legal domains. The benchmark comprises 393 queries verified by a practicing junior advocate with expertise in criminal law, spanning criminal, civil, corporate, and family law—the broadest evaluation scope for Indian legal QA to date.

Our principal empirical findings are:

- LEXAI achieves 95.5 % citation accuracy (CAR) compared to 84.1 % for zero-shot LLM and 61.6 % for naïve RAG.
- The transition classifier reduces outdated-law citations from 82.3 % to 32.6 %—a 60 % relative reduction.
- Answer completeness reaches 72.5 versus 65.0 for naïve RAG ($\Delta = +7.5$ points).
- The learned classifier achieves 98.7 % F1, outperforming the rule-based baseline (F1 = 0.704) and recovering 60 transition cases missed by exact string matching, including *u/s 376E IPC* and *u/s 482 CrPC*.
- Naïve RAG exhibits a hallucination rate of 77.4 % on structured citations; LEXAI reduces this to 1.0 %.

The remainder of the paper is organised as follows. Section II reviews related work. Section III describes the LEXAI architecture. Section ?? presents LEXEVAL-INDIA-393 and evaluation methodology. Section IV reports experimental results and ablation studies. Section V reflects on limitations, and Section VI concludes.

II. RELATED WORK

A. Legal NLP and AI-Assisted Legal Research

Computational approaches to legal text have a long history. Early work focused on statutory interpretation and case retrieval through keyword and citation-graph methods [5]—approaches that remain competitive in narrow retrieval tasks despite the rise of neural methods. Transformer-based models substantially raised the ceiling on legal NLP tasks including case outcome prediction [6], legal judgment prediction [9], and statutory reasoning [10], though gains have not been uniform across jurisdictions and query types.

Domain-specific pre-training has proven important. LegalBERT [7] demonstrated that pre-training on legal corpora meaningfully outperforms general-domain BERT on legal classification tasks. For the Indian jurisdiction, InLegalBERT [8] provided the first pre-trained model on Indian court judgments, achieving strong results on legal named entity recognition and rhetorical role labelling. Both models share a structural limitation: their training corpora predate the 2023 criminal law transition, encoding superseded legislation as current law. This is an inherent property of static pre-training, not a failure of those models—but it illustrates why statutory currency cannot be addressed at the pre-training stage alone.

B. Retrieval-Augmented Generation for Legal QA

The appeal of retrieval-augmented generation [3] for legal applications is straightforward: grounding LLM outputs in retrieved documents reduces the risk of hallucinated citations and fabricated statutory provisions. louis2022statutory demonstrated RAG-based statutory QA on European legislation; nguyen2023 applied RAG to Vietnamese legal documents, reporting that domain-specific retrieval substantially reduces factual errors; and martinez2023legalqa found that hybrid dense-sparse architectures outperform single-modality retrieval on Spanish civil law—a finding consistent with our own ablations.

For Indian legal text, mohan2022 applied extractive QA over Indian Supreme Court judgments but did not address statutory reference or legislative transitions. No prior work, to our knowledge, applies RAG to Indian statutory law in the post-BNS transition period, or treats legislative supersession as an explicit retrieval problem: one where the correct answer depends not only on finding relevant documents but on knowing which of those documents reflects law currently in force.

C. Hallucination and Reliability in Legal AI

Hallucination in legal AI carries consequences that differ in kind from general-purpose domains: an incorrect citation is not merely a factual error but a professional liability [15].

magesh2024 documented systematic hallucination in commercial legal AI products, finding fabricated case citations at rates between 17% and 33% on standardised queries—rates high enough to be professionally untenable. Our evaluation finds the problem is more severe in the Indian post-transition context: the zero-shot LLM exhibits a hallucination rate of 98.3% on structured citations, and naïve RAG reaches 77.4%, underscoring the need for retrieval grounding combined with explicit transition-awareness.

Confidence-based abstention offers one mitigation path. kadavath2022language showed that language models can, to a meaningful degree, distinguish queries they can answer reliably from those they cannot. feng2023knowledge applied this to legal QA, demonstrating that abstaining on low-confidence queries substantially improves precision. LEXAI builds on this with empirically tuned confidence thresholds calibrated on a held-out validation set, making the accuracy-coverage trade-off explicit and adjustable.

D. Legislative Transition and Temporal Legal Reasoning

Tracking which provisions are current, amended, or superseded has received limited attention in legal NLP. gov-ernatori2005temporal formalised temporal legal reasoning in defeasible logic without extending to computational retrieval. palmirani2011akoma proposed structured legislative markup for tracking amendments in European law—an elegant solution that requires substantial manual expert annotation and limits scalability.

For *statutory replacement*, where an entire act supersedes another, no prior computational treatment exists to our knowledge. The closest related work is legislative change detection [21], which identifies textual differences between document pairs but does not classify individual section references as current or superseded at inference time. Our learned transition classifier addresses this gap without requiring structured markup, operating on sentence embeddings from natural language text. The concurrent replacement of IPC, CrPC, and IEA—affecting every criminal proceeding in India—presents an unprecedented test case, and we present both the first computational approach and the first benchmark for evaluating AI systems within it.

III. SYSTEM ARCHITECTURE

A. Overview

LEXAI comprises four main components: a hybrid retrieval pipeline over a four-collection legal database; a learned statutory transition classifier for detecting superseded legislation; a confidence-threshold abstention mechanism; and a structured prompt layer that assembles enriched context for the LLM. At inference time a user query traverses a seven-step pipeline: query classification, parallel hybrid retrieval, confidence filtering, overruling detection, amendment checking, transition enrichment, and LLM generation.

B. Legal Knowledge Base

LEXAI maintains four ChromaDB collections. **Bare Acts** contains 500+ sections from 18 major Indian statutes including both superseded acts (IPC, CrPC, IEA) and their 2023

replacements (BNS, BNSS, BSA), sourced from official Government Gazette PDFs. **Case Law** contains 15,000+ documents comprising Supreme Court judgment chunks augmented with structured question-answer pairs. **Amendments** contains 50+ verified amendment records tracking legislative changes from 2013 to 2023. **Overruling Map** contains 30+ verified Supreme Court overruling relationships, enabling the system to flag when a cited precedent has been subsequently overruled.

C. Hybrid Retrieval Pipeline

Naïve single-modality dense retrieval performs poorly on legal section references because section numbers are sparse tokens that carry disproportionate semantic weight. We address this through a two-stage hybrid pipeline.

Stage 1 — Parallel Retrieval and Fusion. For each query we perform parallel retrieval using dense retrieval via `all-MiniLM-L6-v2` [22] embeddings with cosine similarity, and sparse BM25 retrieval [23] with a legal-domain tokeniser that preserves section references as atomic tokens. Results are merged via Reciprocal Rank Fusion [24]:

$$\text{RRF}(d) = \sum_{r \in R} \frac{1}{k + r(d)} \quad (1)$$

where $r(d)$ is the rank of document d in ranker r , R is the set of rankers, and $k=60$ following `cormack2009reciprocal`. The top-20 candidates proceed to Stage 2.

Stage 2 — Cross-Encoder Reranking. The 20 candidates are reranked using `cross-encoder/ms-marco-MiniLM-L-6-v2` [22], applied separately over the `bare_acts` and `case_law` collections. The top-5 documents by cross-encoder score constitute the final retrieved context.

D. Confidence-Threshold Abstention

After retrieval the system assigns a confidence level based on the cross-encoder similarity score of the top-ranked document:

$$\text{conf}(q) = \begin{cases} \text{HIGH} & \text{if } s_1 \geq \theta_H \\ \text{MEDIUM} & \text{if } \theta_M \leq s_1 < \theta_H \\ \text{LOW} & \text{if } s_1 < \theta_M \end{cases} \quad (2)$$

Thresholds θ_H and θ_M are selected via ablation over six candidate pairs on a 50-query held-out validation set, optimising:

$$\mathcal{O} = 0.45 \cdot \text{Acc} + 0.45 \cdot (1 - \text{HR}) + 0.10 \cdot (1 - \text{AbstRate}) \quad (3)$$

Queries assigned LOW confidence trigger an explicit abstention response, following the selective prediction framework of `feng2023knowledge`. The observed abstention rate of 22.1 % on the full 393-query benchmark reflects this mechanism operating as designed.

E. Learned Statutory Transition Classifier

1) Motivation: A retrieved document may reference a valid legal section that is nonetheless superseded by newer legislation. Prior approaches use manually curated lookup tables mapping old section numbers to new ones. Our rule-based baseline fails on 63 of 234 test cases (26.9 %) due to

surface-form variation: abbreviated references (*u/s 376E IPC*), dotted acronyms (*S. n I.P.C.*), and sections absent from the dictionary (*IPC Section 503*). A classifier operating over sentence embeddings generalises to unseen surface forms without exhaustive enumeration.

2) Dataset Construction: We construct a transition detection dataset of 912 labelled examples. Positive examples (SUPERSEDED) are generated from 118 verified section mappings—83 IPC-to-BNS pairs and 35 CrPC-to-BNSS pairs—derived from official gazette notifications, rendered through eight surface-form templates including formal (*Section {n} of the Indian Penal Code*), abbreviated (*u/s {n} IPC*), dotted (*Sec. {n} I.P.C.*), and procedural (*under Section {n} of the Code of Criminal Procedure*) variants. Negative examples (NOT_SUPERSEDED) are generated from sections of unreplaced acts using the same templates. The dataset is split 80/20 by stratified sampling, yielding 684 training and 234 test examples.

3) Model and Integration: Input section references are encoded using `all-MiniLM-L6-v2`, producing 384-dimensional sentence embeddings. A logistic regression classifier with L2 regularisation is trained on these embeddings. When a reference is classified as SUPERSEDED, the system retrieves the corresponding replacement provision and appends a structured transition note to the LLM context:

$\Rightarrow$ *Transition Note: [IPC Section X] has been replaced by [BNS Section Y] effective 1 July 2024. [Summary of key changes].*

F. LexEval-India-393 Benchmark Dataset

1) Construction: LEXEVAL-INDIA-393 extends the original 293-query criminal law benchmark with 100 additional queries across civil, corporate, and family law, for a total of 393 verified queries. Query generation followed three phases: systematic generation from act text across seven predefined categories with coverage of the BNS transition period; verification by a practicing junior advocate who confirmed citation ground truth and answer summaries; and stratification to ensure balanced category and domain representation.

2) Dataset Statistics: Table I shows the category and domain distribution across the 393 queries.

All 393 queries carry citation ground truth (`correct_citation`) and verified answer summaries (`correct_answer_summary`) confirmed by the legal expert. BNS transition applies to 55 queries (14.0 %).

A note on ACS scoring methodology. Answer completeness for criminal law queries (293 queries) uses a six-component rubric scorer calibrated during initial system design, which evaluates completeness across dimensions including citation accuracy, statutory detail, and procedural context. Cross-domain queries (100 queries: Civil, Corporate, Family) use semantic similarity against `correct_answer_summary` via `paraphrase-multilingual-MiniLM-L12-v2`, applied where answer summaries follow a deterministic bare-act structure. This hybrid approach was adopted to handle structural

TABLE I
LEXEVAL-INDIA-393 DISTRIBUTION BY CATEGORY AND DOMAIN

Category	Domain	N	%
Section Lookup	All	143	36.4
IPC to BNS Transition	Criminal	50	12.7
Case Law Search	Criminal	50	12.7
Punishment Queries	Criminal	40	10.2
Amendment Specific	Criminal	40	10.2
Complex Interpretation	Criminal	40	10.2
Overruled Case Detection	Criminal	30	7.6
<i>By Domain</i>			
	Criminal Law	293	74.6
	Civil Law	40	10.2
	Corporate Law	30	7.6
	Family Law	30	7.6
Total		393	100

differences between criminal law queries—which include case-law-heavy and doctrinal questions—and the predominantly section-lookup nature of cross-domain queries. ACS figures across domains should be interpreted with this methodological distinction in mind.

IV. EXPERIMENTS

A. Baselines and Evaluation Metrics

We compare LEXAI against two baselines: **NoRAG**, a zero-shot LLM with no retrieval grounding; and **SimpleRAG**, a naïve RAG system using dense cosine-similarity retrieval without reranking, transition detection, or confidence abstention.

We report five metrics. **CAR** (Citation Accuracy Rate, %) measures the fraction of generated citations matching lawyer-verified ground truth. **HR** (Hallucination Rate, %) measures structured citations to section numbers absent from the legal database. **OLR** (Outdated-Law Rate, %) measures citations to provisions superseded by the 2023 transition. **ACS** (Answer Completeness Score) measures answer completeness against ground-truth summaries (see scoring note in Section ??). **AP** (Abstention Performance) measures F1 of the confidence-threshold abstention mechanism.

B. Main Results

Table II presents the evaluation results across all three systems on LEXEVAL-INDIA-393.

Citation Accuracy (CAR). LEXAI achieves 95.6 % citation accuracy, ahead of NoRAG by 11.4 pp and SimpleRAG by 34.0 pp. The gap against SimpleRAG is striking: naïve dense retrieval treats section numbers as low-weight tokens, causing the retriever to surface topically relevant but section-incorrect documents. Hybrid retrieval with cross-encoder reranking substantially corrects this, while the transition classifier additionally steers the LLM toward current provisions. NoRAG’s relatively high CAR of 84.1 % reflects that many queries involve sections whose basic identity the LLM has memorised from pre-training—but this comes at the cost of nearly universal hallucination, as the following metrics show.

Hallucination Rate (HR). The hallucination figures reveal the severity of the problem this work addresses. NoRAG fabricates structured citations on 98.3 % of queries—essentially every response contains at least one citation to a section number absent from the legal database. SimpleRAG reduces this to 77.4 % but remains professionally untenable. LEXAI achieves 1.0 %, a reduction of 97.3 pp over NoRAG and 76.4 pp over SimpleRAG. The residual 1.0 % hallucination warrants attention: it corresponds to approximately four queries out of 393, and arises from cases where the LLM generates a citation not present in retrieved context, suggesting that retrieval grounding alone does not entirely prevent parametric confabulation.

Outdated-Law Rate (OLR). NoRAG cites superseded law on 82.3 % of queries. LEXAI reduces this to 32.6 %, a 60 % relative reduction attributable primarily to the transition classifier. The residual 32.6 % represents cases where the classifier did not flag the query—either because the reference is indirect, the provision has no clean BNS equivalent, or the framing fell outside the classifier’s training distribution. SimpleRAG reduces OLR to 48.8 % through retrieval grounding alone, confirming that both retrieval quality and explicit transition detection contribute independently.

Answer Completeness (ACS). LEXAI achieves an ACS of 72.5 versus 65.0 for SimpleRAG (+7.5 points) and 72.2 for NoRAG (+0.3 points). The improvement over SimpleRAG reflects the richer context assembled by the full pipeline—transition notes, amendment annotations, and overruling flags contribute answer components that naïve retrieval cannot provide. The small gap relative to NoRAG is consistent with LLMs having broad pre-trained coverage of common legal questions even when their citations are wrong. Readers should note the scoring methodology distinction described in Section ??: criminal law ACS uses rubric scoring while cross-domain ACS uses semantic similarity, meaning aggregate ACS figures blend two measurement approaches.

Abstention Performance (AP). LEXAI achieves AP F1 of 0.628 at an abstention rate of 22.1 %, with perfect abstention precision (1.000) and recall of 0.458. This indicates the confidence mechanism abstains conservatively: when it declines to answer it is always correct to do so, though it misses approximately half of queries where abstention would be appropriate. SimpleRAG’s AP F1 of 0.707 at an abstention rate of 38.4 % reflects a more aggressive abstention policy, trading coverage for precision in a different operating regime. NoRAG, with no abstention mechanism, achieves AP F1 of 0.000 by construction.

C. Performance by Query Category

LEXAI performs most reliably on Section Lookup and IPC-to-BNS Transition queries, where ground truth is unambiguous and retrieving the correct document is generally sufficient. Performance on Complex Legal Interpretation queries is lower, consistent with the inherent difficulty of open-ended legal reasoning for retrieval-based systems. The strong performance on IPC-to-BNS Transition queries provides evidence that the

TABLE II

MAIN EVALUATION RESULTS ON LEXEVAL-INDIA-393 (393 QUERIES ACROSS 4 DOMAINS). ALL METRICS IN %. HIGHER CAR AND ACS ARE BETTER ($\uparrow$); LOWER HR AND OLR ARE BETTER ($\downarrow$).

System	CAR $\uparrow$	HR $\downarrow$	OLR $\downarrow$	ACS $\uparrow$	AP F1 $\uparrow$
NoRAG	84.10	98.26	82.25	72.22	0.000
SimpleRAG	61.58	77.39	48.84	65.04	0.707
LexAI (Ours)	95.55	1.00	32.57	72.54	0.628
<i>LexAI Abstention Rate: 22.1% P@1: 2.799 P@3: 2.290 CCS Calibration Error: 0.260</i>					

transition classifier’s gains propagate through to end-to-end answer quality.

D. Cross-Domain Evaluation

Table III shows performance across the four legal domains in LEXEVAL-INDIA-393. The system generalises reasonably across domains despite being trained and primarily evaluated on criminal law.

TABLE III
LEXAI CROSS-DOMAIN PERFORMANCE ON LEXEVAL-INDIA-393

Domain	N	CAR (%)	ACS	<i>Note: Cross-domain</i>
Criminal Law	293	95.55	72.54	
Civil Law	40	<i>see note</i>		
Corporate Law	30	<i>see note</i>		
Family Law	30	<i>see note</i>		
Overall	393	95.55	72.54	

per-category CAR figures are pending re-evaluation under a unified scoring protocol. Section V discusses the methodological reason for this.

E. Ablation Studies

1) *Transition Detection: Learned vs. Rule-Based:* Table IV compares the learned transition classifier against the rule-based dictionary baseline on the 234-example held-out test set.

TABLE IV

TRANSITION CLASSIFIER ABLATION. RULE-BASED: MANUALLY CURATED IPC-TO-BNS DICTIONARY. LEARNED: MINILM + LOGISTIC REGRESSION. TEST SET: 234 EXAMPLES, 80/20 STRATIFIED SPLIT (TRAINING SET: 684 EXAMPLES).

Metric	Rule-Based	Learned (Ours)
Accuracy	0.7237	0.9868
Precision (macro)	0.8055	0.9869
Recall (macro)	0.7237	0.9868
F1 (macro)	0.7039	0.9868
Error Cases	63	3
Inference (s)	0.188	0.440

The learned classifier correctly classifies 231 of 234 test examples, recovering 60 cases where the dictionary fails. Error analysis reveals that rule-based failures cluster around abbreviated forms (*u/s 376E IPC*, *u/s 482 CrPC*) and sections absent from the manual dictionary (*IPC Section 503*, *IPC Section 381*). The learned classifier handles these by operating over sentence embeddings rather than character sequences, capturing the semantic pattern of a superseded-law reference

without surface-form sensitivity. Three residual errors by the learned classifier—sections that appear in genuinely ambiguous contexts—represent an irreducible minimum at current training scale.

The inference time difference (0.440 s vs. 0.188 s) is modest and acceptable in legal research contexts where accuracy takes priority over low latency.

2) *BNS Middleware Impact on OLR:* Disabling the transition middleware causes LEXAI’s OLR on IPC-to-BNS transition queries to rise substantially, confirming that OLR improvement is attributable to the transition module specifically rather than to retrieval quality gains shared with SimpleRAG. The transition module is therefore a decomposable, independently verifiable contribution.

3) *Confidence Threshold Sensitivity:* The empirically selected thresholds (θ_H , θ_M) achieve the highest weighted objective score among six candidate pairs. Performance degrades predictably in both directions: very high thresholds produce excessive abstention and reduced coverage, while very low thresholds admit low-confidence answers that hurt precision. The optimal operating point is reasonably stable across modest distributional shifts, implying the system does not require continuous recalibration in production.

V. LIMITATIONS

ACS Scoring Heterogeneity. The most significant methodological limitation of this evaluation is the use of two different ACS scoring approaches: a six-component rubric scorer for criminal law queries (293 queries) and semantic similarity scoring for cross-domain queries (100 queries). These produce scores on different effective scales, meaning aggregate ACS figures blend two measurement approaches. This decision was made to accommodate structural differences between criminal law answers—which involve case law, procedural rules, and doctrinal analysis—and the predominantly bare-act answers of cross-domain queries. Future work should unify the scoring methodology or report domain-specific ACS separately with clearly labelled methods.

Answer Summary Quality. Inspection of the ground truth Excel reveals that `correct_answer_summary` fields follow a templated structure (“Section X of the IPC is a statutory provision under Indian criminal law that defines the legal rule...”) rather than substantive legal summaries. This means ACS, which compares generated answers against these summaries, partially measures surface similarity to a template rather than substantive legal correctness. This is a limitation

of the current ground truth and should be addressed in future benchmark iterations with more specific answer summaries authored or reviewed by legal experts.

Residual OLR. The 32.6 % residual outdated-law rate, while far below the 82.3 % baseline, is not negligible. It reflects queries where the classifier did not flag a superseded reference—whether because the reference is indirect, the provision has no clean BNS equivalent, or the framing falls outside the classifier’s training distribution. Understanding and reducing this residual is an important direction for future work.

Classifier Test Set Distribution. The learned transition classifier achieves 98.7 % F1 on the held-out test set. While the split was stratified and held out before training, the test set derives from the same surface-form templates as the training data. Performance on genuinely novel section references from practitioner queries—rather than generated templates—has not been confirmed and may differ.

Cross-Domain CAR Reporting. The cross-domain evaluation (Civil, Corporate, Family Law) shares the CAR metric definition but the 100 additional queries were evaluated under conditions that may differ from the 293 criminal law queries, including different ACS scoring. Per-domain CAR figures broken down by system (LexAI vs. baselines) are not reported here and should be added in a camera-ready version.

English Only. All queries and database content are in English. Indian court proceedings routinely incorporate Hindi and regional language terms. Extending retrieval and generation to multilingual settings is an important practical direction.

Criminal Law Bias. LEXEVAL-INDIA-393 reflects the primary practice area of our legal expert consultant. The 100 cross-domain queries are a preliminary extension; a full evaluation on civil, corporate, and family law warrants a dedicated benchmark with equivalent depth.

VI. CONCLUSION

We have described LEXAI, a RAG system for Indian legal research built around statutory reliability as a primary design constraint. The central contribution is a learned transition classifier that detects references to superseded legislation in retrieved context and enriches LLM input with the corresponding current provisions. Where a manually curated dictionary fails on 63 of 234 test cases, the learned classifier fails on only 3, recovering 60 transition cases including abbreviated and vernacular section references common in Indian legal practice.

Evaluated on LEXEVAL-INDIA-393—393 queries across criminal, civil, corporate, and family law—LEXAI achieves 95.6 % citation accuracy versus 84.1 % for zero-shot LLM and 61.6 % for naïve RAG. Critically, the hallucination rate drops from 98.3 % (NoRAG) and 77.4 % (SimpleRAG) to 1.0 %—a reduction that matters in professional contexts where a single fabricated citation can cause procedural harm. The transition module reduces outdated-law citations from 82.3 % to 32.6 %, and answer completeness improves by 7.5 points over naïve RAG.

Beyond the Indian context, the work points toward a general approach to legislative transitions in legal AI: statutory obsoles-

cence can be framed as a classification problem over sentence embeddings, operating at inference time without structured markup or ongoing dictionary maintenance. As jurisdictions worldwide replace colonial-era statutes with contemporary legislation, classifiers trained on transition mappings offer a practical path to maintaining AI research reliability through transitional periods.

We release LEXEVAL-INDIA-393 and all evaluation code publicly, in the hope that the benchmark and methodology prove useful to researchers working on Indian legal NLP and, more broadly, on the problem of keeping legal AI systems aligned with the law as it actually stands.

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